

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 11A

FURTHER KINETICS (RATES, ORDERS, MECHANISMS & ARRHENIUS)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 11: KINETICS 2
SUB-TOPIC:	11A: Further Kinetics (Rates, Orders, Mechanisms & Arrhenius)
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	202 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. Continuous Monitoring of Reaction Rate****[4 Marks]**
WCH14/01/Jan23/Q1

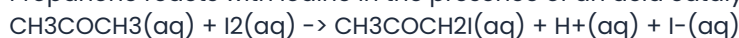
The reaction between magnesium ribbon and dilute hydrochloric acid was investigated:



- (a) State the most suitable experimental method to continuously monitor the rate of this reaction and explain your choice. [2]
- (b) Sketch a graph of volume of hydrogen gas produced against time, labelling the axes and indicating how initial rate is determined. [2]

Q2. 2. Colorimetric Measurement of Reaction Rate**[3 Marks]**
WCH14/01/Oct22/Q2

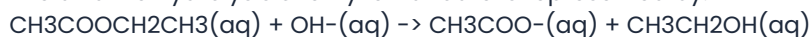
Propanone reacts with iodine in the presence of an acid catalyst:



- (a) Explain how colorimetry can be used to follow the progress of this reaction. [2]
- (b) State why a colorimeter is preferable to titrating samples with sodium thiosulfate in this experiment. [1]

Q3. 3. Quenching and Titrimetry**[4 Marks]**
WCH14/01/Jun22/Q3

The alkaline hydrolysis of ethyl ethanoate is represented by:



- (a) Explain why quenching is necessary before titrating samples taken from this reaction mixture. [2]
- (b) Suggest a suitable quenching method for this reaction. [2]

Q4. 4. Electrical Conductivity Method**[3 Marks]**
WCH14/01/Jan22/Q4

The bromination of propanone produces H^+ and Br^- ions:



- (a) Explain why the electrical conductivity of the mixture increases as the reaction proceeds. [2]
- (b) State how conductivity data can be converted into rate data. [1]

Q5. 5. Rate Equation and Rate Constant Units**[3 Marks]**
WCH14/01/Oct21/Q1

A reaction $\text{A} + 2\text{B} \rightarrow \text{C}$ has the rate equation: $\text{rate} = k[\text{A}][\text{B}]^2$

- (a) State the order of reaction with respect to A, B, and the overall order. [2]
- (b) Deduce the units of the rate constant, k . [1]

Q6. 6. Zero Order Kinetics Definition**[2 Marks]**
WCH14/01/Jun21/Q2

The rate equation for a reaction $\text{X} \rightarrow \text{Y}$ is $\text{rate} = k[\text{X}]^0$

- (a) Explain what zero order with respect to reactant X implies about the reaction mechanism. [1]
- (b) Sketch the concentration–time graph for a zero-order reaction. [1]

Q7. 7. First Order Half-Life Relationship**[3 Marks]**
WCH14/01/Jan21/Q3

The decomposition of hydrogen peroxide $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ is first order.

- (a) Define half-life ($t_{1/2}$) for a chemical reaction. [1]
- (b) Explain how the concentration–time graph confirms that a reaction is first order. [2]

Q8. 8. Half-Life Calculation of k**[3 Marks]**
WCH14/01/Oct20/Q4

For a first-order reaction, the half-life $t_{1/2}$ is 450 seconds.

- (a) Calculate the rate constant, k , using the expression $t_{1/2} = \ln 2 / k$. Include units. [2]
(b) Calculate the rate of reaction when $[\text{reactant}] = 0.080 \text{ mol dm}^{-3}$. [1]

Q9. 9. Initial Rates Table Method 1**[4 Marks]**
WCH14/01/Jun19/Q5

Initial rate data for $P + Q \rightarrow R$:

Exp 1: $[P] = 0.10$, $[Q] = 0.10$, Rate = 1.2×10^{-3}

Exp 2: $[P] = 0.20$, $[Q] = 0.10$, Rate = 2.4×10^{-3}

Exp 3: $[P] = 0.10$, $[Q] = 0.30$, Rate = 1.08×10^{-2}

- (a) Deduce the order of reaction with respect to P and Q. Show your reasoning. [3]
(b) Write the rate equation. [1]

Q10. 10. Calculation of Rate Constant from Data Table**[3 Marks]**
WCH14/01/Jan19/Q6

Using the data from Q9 (Exp 1: $[P] = 0.10 \text{ mol dm}^{-3}$, $[Q] = 0.10 \text{ mol dm}^{-3}$, Rate = $1.2 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$):

- (a) Calculate the value of the rate constant, k . [2]
(b) State the units of k . [1]

Q11. 11. Kinetic Analysis & Mechanism Step 11**[3 Marks]**
WCH14/01/Sample/Q11

Consider the multi-step reaction mechanism for reaction scheme 11:

Step 1: $A + B \rightarrow AB$ (slow)

Step 2: $AB + B \rightarrow AB_2$ (fast)

- (a) Identify the rate-determining step and explain your choice. [2]
(b) Deduce the rate equation consistent with this mechanism. [1]

Q12. 12. Kinetic Analysis & Mechanism Step 12**[3 Marks]**
WCH14/01/Sample/Q12

Consider the multi-step reaction mechanism for reaction scheme 12:

Step 1: $A + B \rightarrow AB$ (slow)

Step 2: $AB + B \rightarrow AB_2$ (fast)

- (a) Identify the rate-determining step and explain your choice. [2]
(b) Deduce the rate equation consistent with this mechanism. [1]

Q13. 13. Kinetic Analysis & Mechanism Step 13**[3 Marks]**
WCH14/01/Sample/Q13

Consider the multi-step reaction mechanism for reaction scheme 13:

Step 1: $A + B \rightarrow AB$ (slow)

Step 2: $AB + B \rightarrow AB_2$ (fast)

- (a) Identify the rate-determining step and explain your choice. [2]
(b) Deduce the rate equation consistent with this mechanism. [1]

Q14. 14. Kinetic Analysis & Mechanism Step 14**[3 Marks]**
WCH14/01/Sample/Q14

Consider the multi-step reaction mechanism for reaction scheme 14:

Step 1: $A + B \rightarrow AB$ (slow)

Step 2: $AB + B \rightarrow AB_2$ (fast)

- (a) Identify the rate-determining step and explain your choice. [2]
(b) Deduce the rate equation consistent with this mechanism. [1]

Q15. 15. Kinetic Analysis & Mechanism Step 15**[3 Marks]****WCH14/01/Sample/Q15**

Consider the multi-step reaction mechanism for reaction scheme 15:

Step 1: $A + B \rightarrow AB$ (slow)

Step 2: $AB + B \rightarrow AB_2$ (fast)

- (a) Identify the rate-determining step and explain your choice. [2]
(b) Deduce the rate equation consistent with this mechanism. [1]

Q16. 16. Kinetic Analysis & Mechanism Step 16**[3 Marks]****WCH14/01/Sample/Q16**

Consider the multi-step reaction mechanism for reaction scheme 16:

Step 1: $A + B \rightarrow AB$ (slow)

Step 2: $AB + B \rightarrow AB_2$ (fast)

- (a) Identify the rate-determining step and explain your choice. [2]
(b) Deduce the rate equation consistent with this mechanism. [1]

Q17. 17. Kinetic Analysis & Mechanism Step 17**[3 Marks]****WCH14/01/Sample/Q17**

Consider the multi-step reaction mechanism for reaction scheme 17:

Step 1: $A + B \rightarrow AB$ (slow)

Step 2: $AB + B \rightarrow AB_2$ (fast)

- (a) Identify the rate-determining step and explain your choice. [2]
(b) Deduce the rate equation consistent with this mechanism. [1]

Q18. 18. Kinetic Analysis & Mechanism Step 18**[3 Marks]****WCH14/01/Sample/Q18**

Consider the multi-step reaction mechanism for reaction scheme 18:

Step 1: $A + B \rightarrow AB$ (slow)

Step 2: $AB + B \rightarrow AB_2$ (fast)

- (a) Identify the rate-determining step and explain your choice. [2]
(b) Deduce the rate equation consistent with this mechanism. [1]

Q19. 19. Kinetic Analysis & Mechanism Step 19**[3 Marks]****WCH14/01/Sample/Q19**

Consider the multi-step reaction mechanism for reaction scheme 19:

Step 1: $A + B \rightarrow AB$ (slow)

Step 2: $AB + B \rightarrow AB_2$ (fast)

- (a) Identify the rate-determining step and explain your choice. [2]
(b) Deduce the rate equation consistent with this mechanism. [1]

Q20. 20. Kinetic Analysis & Mechanism Step 20**[3 Marks]****WCH14/01/Sample/Q20**

Consider the multi-step reaction mechanism for reaction scheme 20:

Step 1: $A + B \rightarrow AB$ (slow)

Step 2: $AB + B \rightarrow AB_2$ (fast)

- (a) Identify the rate-determining step and explain your choice. [2]
(b) Deduce the rate equation consistent with this mechanism. [1]

Q21. 21. Kinetic Analysis & Mechanism Step 21**[3 Marks]****WCH14/01/Sample/Q21**

Consider the multi-step reaction mechanism for reaction scheme 21:

Step 1: $A + B \rightarrow AB$ (slow)

Step 2: $AB + B \rightarrow AB_2$ (fast)

- (a) Identify the rate-determining step and explain your choice. [2]
- (b) Deduce the rate equation consistent with this mechanism. [1]

Q22. 22. Kinetic Analysis & Mechanism Step 22**[3 Marks]****WCH14/01/Sample/Q22**

Consider the multi-step reaction mechanism for reaction scheme 22:

Step 1: $A + B \rightarrow AB$ (slow)

Step 2: $AB + B \rightarrow AB_2$ (fast)

- (a) Identify the rate-determining step and explain your choice. [2]
- (b) Deduce the rate equation consistent with this mechanism. [1]

Q23. 23. Kinetic Analysis & Mechanism Step 23**[3 Marks]****WCH14/01/Sample/Q23**

Consider the multi-step reaction mechanism for reaction scheme 23:

Step 1: $A + B \rightarrow AB$ (slow)

Step 2: $AB + B \rightarrow AB_2$ (fast)

- (a) Identify the rate-determining step and explain your choice. [2]
- (b) Deduce the rate equation consistent with this mechanism. [1]

Q24. 24. Kinetic Analysis & Mechanism Step 24**[3 Marks]****WCH14/01/Sample/Q24**

Consider the multi-step reaction mechanism for reaction scheme 24:

Step 1: $A + B \rightarrow AB$ (slow)

Step 2: $AB + B \rightarrow AB_2$ (fast)

- (a) Identify the rate-determining step and explain your choice. [2]
- (b) Deduce the rate equation consistent with this mechanism. [1]

Q25. 25. Kinetic Analysis & Mechanism Step 25**[3 Marks]****WCH14/01/Sample/Q25**

Consider the multi-step reaction mechanism for reaction scheme 25:

Step 1: $A + B \rightarrow AB$ (slow)

Step 2: $AB + B \rightarrow AB_2$ (fast)

- (a) Identify the rate-determining step and explain your choice. [2]
- (b) Deduce the rate equation consistent with this mechanism. [1]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. A* Challenge: Arrhenius & RDS Mechanism Analysis****[5 Marks]****26****WCH14/01/Hard/Q26**

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .

[4]

- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q27. 27. A* Challenge: Arrhenius & RDS Mechanism Analysis**[5 Marks]****27****WCH14/01/Hard/Q27**

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .

[4]

- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q28. 28. A* Challenge: Arrhenius & RDS Mechanism Analysis**[5 Marks]****28****WCH14/01/Hard/Q28**

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .

[4]

- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q29. 29. A* Challenge: Arrhenius & RDS Mechanism Analysis**[5 Marks]****29****WCH14/01/Hard/Q29**

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .

[4]

- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q30. 30. A* Challenge: Arrhenius & RDS Mechanism Analysis**[5 Marks]****30****WCH14/01/Hard/Q30**

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .

[4]

- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q31. 31. A* Challenge: Arrhenius & RDS Mechanism Analysis 31**[5 Marks]****WCH14/01/Hard/Q31**

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} . [4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q32. 32. A* Challenge: Arrhenius & RDS Mechanism Analysis 32**[5 Marks]****WCH14/01/Hard/Q32**

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} . [4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q33. 33. A* Challenge: Arrhenius & RDS Mechanism Analysis 33**[5 Marks]****WCH14/01/Hard/Q33**

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} . [4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q34. 34. A* Challenge: Arrhenius & RDS Mechanism Analysis 34**[5 Marks]****WCH14/01/Hard/Q34**

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} . [4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q35. 35. A* Challenge: Arrhenius & RDS Mechanism Analysis 35**[5 Marks]****WCH14/01/Hard/Q35**

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} . [4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q36. 36. A* Challenge: Arrhenius & RDS Mechanism Analysis**36****[5 Marks]****WCH14/01/Hard/Q36**

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .
[4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q37. 37. A* Challenge: Arrhenius & RDS Mechanism Analysis**37****[5 Marks]****WCH14/01/Hard/Q37**

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .
[4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q38. 38. A* Challenge: Arrhenius & RDS Mechanism Analysis**38****[5 Marks]****WCH14/01/Hard/Q38**

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .
[4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q39. 39. A* Challenge: Arrhenius & RDS Mechanism Analysis**39****[5 Marks]****WCH14/01/Hard/Q39**

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .
[4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q40. 40. A* Challenge: Arrhenius & RDS Mechanism Analysis**40****[5 Marks]****WCH14/01/Hard/Q40**

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .
[4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q41. 41. A* Challenge: Arrhenius & RDS Mechanism Analysis 41**[5 Marks]**
WCH14/01/Hard/Q41

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .
[4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q42. 42. A* Challenge: Arrhenius & RDS Mechanism Analysis 42**[5 Marks]**
WCH14/01/Hard/Q42

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .
[4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q43. 43. A* Challenge: Arrhenius & RDS Mechanism Analysis 43**[5 Marks]**
WCH14/01/Hard/Q43

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .
[4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q44. 44. A* Challenge: Arrhenius & RDS Mechanism Analysis 44**[5 Marks]**
WCH14/01/Hard/Q44

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .
[4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q45. 45. A* Challenge: Arrhenius & RDS Mechanism Analysis 45**[5 Marks]**
WCH14/01/Hard/Q45

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .
[4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q46. 46. A* Challenge: Arrhenius & RDS Mechanism Analysis
46**[5 Marks]**
WCH14/01/Hard/Q46

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .
[4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q47. 47. A* Challenge: Arrhenius & RDS Mechanism Analysis
47**[5 Marks]**
WCH14/01/Hard/Q47

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .
[4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q48. 48. A* Challenge: Arrhenius & RDS Mechanism Analysis
48**[5 Marks]**
WCH14/01/Hard/Q48

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .
[4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q49. 49. A* Challenge: Arrhenius & RDS Mechanism Analysis
49**[5 Marks]**
WCH14/01/Hard/Q49

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .
[4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

Q50. 50. A* Challenge: Arrhenius & RDS Mechanism Analysis
50**[5 Marks]**
WCH14/01/Hard/Q50

An investigation into the temperature dependence of the rate constant k yielded the following data:

At $T = 300\text{ K}$, $k = 2.0 \times 10^{-4}\text{ s}^{-1}$.

At $T = 350\text{ K}$, $k = 1.5 \times 10^{-2}\text{ s}^{-1}$.

(Gas constant $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$)

- (a) Use the Arrhenius equation $\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$ to calculate the activation energy, E_a , in kJ mol^{-1} .
[4]
- (b) Explain the effect of adding a heterogeneous catalyst on the value of E_a and the rate constant k . [1]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 11A

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Continuous Monitoring of Reaction Rate 1. (a) Measure gas volume using a gas syringe / measuring cylinder over water (1). Hydrogen is a gas produced while reactants are solid/liquid (1). 1. (b) Curve starting at origin, steep initial gradient levelling off to horizontal (1). Draw tangent at $t=0$; initial rate = gradient of tangent (1).	[4]
Q2	Q2: 2. Colorimetric Measurement of Reaction Rate 2. (a) Iodine is brown/yellow while all other species are colourless (1). Measure absorbance over time using a colorimeter (1). 2. (b) Colorimetry allows continuous monitoring without stopping/quenching the reaction (1).	[3]
Q3	Q3: 3. Quenching and Titrimetry 3. (a) Quenching rapidly stops or slows down the reaction (1) so the concentration of OH^- does not change during titration (1). 3. (b) Add sample to ice-cold water (1) or add a known excess of ice-cold acid (1).	[4]
Q4	Q4: 4. Electrical Conductivity Method 4. (a) Total concentration of ions increases (from 1 H^+ to 2 H^+ and 1 Br^-) (1); H^+ ions have high ionic conductivity (1). 4. (b) Plot conductivity vs time and draw tangent at $t=0$ (1).	[3]
Q5	Q5: 5. Rate Equation and Rate Constant Units 5. (a) Order w.r.t A = 1, w.r.t B = 2 (1); Overall order = $1 + 2 = 3$ (1). 5. (b) units of $k = (\text{mol dm}^{-3} \text{ s}^{-1}) / ((\text{mol dm}^{-3})(\text{mol dm}^{-3})^2) = \text{dm}^6 \text{ mol}^{-2} \text{ s}^{-1}$ (1).	[3]
Q6	Q6: 6. Zero Order Kinetics Definition 6. (a) X is not involved in the rate-determining step (or RDS occurs after X acts) (1). 6. (b) Straight line with negative gradient (1).	[2]
Q7	Q7: 7. First Order Half-Life Relationship 7. (a) Time taken for the concentration of a reactant to decrease to half of its initial value (1). 7. (b) Half-life is constant (1) independent of initial concentration (1).	[3]
Q8	Q8: 8. Half-Life Calculation of k 8. (a) $k = 0.693 / 450 = 1.54 \times 10^{-3} \text{ s}^{-1}$ (2). 8. (b) rate = $(1.54 \times 10^{-3})(0.080) = 1.23 \times 10^{-4} \text{ mol dm}^{-3} \text{ s}^{-1}$ (1).	[3]
Q9	Q9: 9. Initial Rates Table Method 1 9. (a) Exp 1 $\rightarrow$ 2: $[\text{P}] \times 2$, Rate $\times 2 \rightarrow$ 1st order w.r.t P (1). Exp 1 $\rightarrow$ 3: $[\text{Q}] \times 3$, Rate $\times 9$ (3^2) $\rightarrow$ 2nd order w.r.t Q (2). 9. (b) rate = $k[\text{P}][\text{Q}]^2$ (1).	[4]
Q10	Q10: 10. Calculation of Rate Constant from Data Table 10. (a) $k = 1.2 \times 10^{-3} / ((0.10)(0.10)^2) = 1.20 \text{ dm}^6 \text{ mol}^{-2} \text{ s}^{-1}$ (2). 10. (b) $\text{dm}^6 \text{ mol}^{-2} \text{ s}^{-1}$ (1).	[3]
Q11	Q11: 11. Kinetic Analysis & Mechanism Step 11 11. (a) Step 1 is the rate-determining step (1) because it is the slowest step in the mechanism (1). 11. (b) rate = $k[\text{A}][\text{B}]$ (1).	[3]
Q12	Q12: 12. Kinetic Analysis & Mechanism Step 12 12. (a) Step 1 is the rate-determining step (1) because it is the slowest step in the mechanism (1). 12. (b) rate = $k[\text{A}][\text{B}]$ (1).	[3]
Q13	Q13: 13. Kinetic Analysis & Mechanism Step 13 13. (a) Step 1 is the rate-determining step (1) because it is the slowest step in the mechanism (1). 13. (b) rate = $k[\text{A}][\text{B}]$ (1).	[3]

Q14	Q14: 14. Kinetic Analysis & Mechanism Step 14 14. (a) Step 1 is the rate-determining step (1) because it is the slowest step in the mechanism (1). 14. (b) rate = $k[A][B]$ (1).	[3]
Q15	Q15: 15. Kinetic Analysis & Mechanism Step 15 15. (a) Step 1 is the rate-determining step (1) because it is the slowest step in the mechanism (1). 15. (b) rate = $k[A][B]$ (1).	[3]
Q16	Q16: 16. Kinetic Analysis & Mechanism Step 16 16. (a) Step 1 is the rate-determining step (1) because it is the slowest step in the mechanism (1). 16. (b) rate = $k[A][B]$ (1).	[3]
Q17	Q17: 17. Kinetic Analysis & Mechanism Step 17 17. (a) Step 1 is the rate-determining step (1) because it is the slowest step in the mechanism (1). 17. (b) rate = $k[A][B]$ (1).	[3]
Q18	Q18: 18. Kinetic Analysis & Mechanism Step 18 18. (a) Step 1 is the rate-determining step (1) because it is the slowest step in the mechanism (1). 18. (b) rate = $k[A][B]$ (1).	[3]
Q19	Q19: 19. Kinetic Analysis & Mechanism Step 19 19. (a) Step 1 is the rate-determining step (1) because it is the slowest step in the mechanism (1). 19. (b) rate = $k[A][B]$ (1).	[3]
Q20	Q20: 20. Kinetic Analysis & Mechanism Step 20 20. (a) Step 1 is the rate-determining step (1) because it is the slowest step in the mechanism (1). 20. (b) rate = $k[A][B]$ (1).	[3]
Q21	Q21: 21. Kinetic Analysis & Mechanism Step 21 21. (a) Step 1 is the rate-determining step (1) because it is the slowest step in the mechanism (1). 21. (b) rate = $k[A][B]$ (1).	[3]
Q22	Q22: 22. Kinetic Analysis & Mechanism Step 22 22. (a) Step 1 is the rate-determining step (1) because it is the slowest step in the mechanism (1). 22. (b) rate = $k[A][B]$ (1).	[3]
Q23	Q23: 23. Kinetic Analysis & Mechanism Step 23 23. (a) Step 1 is the rate-determining step (1) because it is the slowest step in the mechanism (1). 23. (b) rate = $k[A][B]$ (1).	[3]
Q24	Q24: 24. Kinetic Analysis & Mechanism Step 24 24. (a) Step 1 is the rate-determining step (1) because it is the slowest step in the mechanism (1). 24. (b) rate = $k[A][B]$ (1).	[3]
Q25	Q25: 25. Kinetic Analysis & Mechanism Step 25 25. (a) Step 1 is the rate-determining step (1) because it is the slowest step in the mechanism (1). 25. (b) rate = $k[A][B]$ (1).	[3]
Q26	Q26: 26. A* Challenge: Arrhenius & RDS Mechanism Analysis 26 26. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 26. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]

Q27	Q27: 27. A* Challenge: Arrhenius & RDS Mechanism Analysis 27 27. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 27. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q28	Q28: 28. A* Challenge: Arrhenius & RDS Mechanism Analysis 28 28. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 28. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q29	Q29: 29. A* Challenge: Arrhenius & RDS Mechanism Analysis 29 29. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 29. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q30	Q30: 30. A* Challenge: Arrhenius & RDS Mechanism Analysis 30 30. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 30. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q31	Q31: 31. A* Challenge: Arrhenius & RDS Mechanism Analysis 31 31. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 31. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q32	Q32: 32. A* Challenge: Arrhenius & RDS Mechanism Analysis 32 32. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 32. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q33	Q33: 33. A* Challenge: Arrhenius & RDS Mechanism Analysis 33 33. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 33. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q34	Q34: 34. A* Challenge: Arrhenius & RDS Mechanism Analysis 34 34. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 34. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q35	Q35: 35. A* Challenge: Arrhenius & RDS Mechanism Analysis 35 35. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 35. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q36	Q36: 36. A* Challenge: Arrhenius & RDS Mechanism Analysis 36 36. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 36. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q37	Q37: 37. A* Challenge: Arrhenius & RDS Mechanism Analysis 37 37. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 37. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q38	Q38: 38. A* Challenge: Arrhenius & RDS Mechanism Analysis 38 38. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 38. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q39	Q39: 39. A* Challenge: Arrhenius & RDS Mechanism Analysis 39 39. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 39. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]

Q40	Q40: 40. A* Challenge: Arrhenius & RDS Mechanism Analysis 40 40. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 40. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q41	Q41: 41. A* Challenge: Arrhenius & RDS Mechanism Analysis 41 41. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 41. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q42	Q42: 42. A* Challenge: Arrhenius & RDS Mechanism Analysis 42 42. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 42. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q43	Q43: 43. A* Challenge: Arrhenius & RDS Mechanism Analysis 43 43. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 43. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q44	Q44: 44. A* Challenge: Arrhenius & RDS Mechanism Analysis 44 44. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 44. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q45	Q45: 45. A* Challenge: Arrhenius & RDS Mechanism Analysis 45 45. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 45. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q46	Q46: 46. A* Challenge: Arrhenius & RDS Mechanism Analysis 46 46. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 46. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q47	Q47: 47. A* Challenge: Arrhenius & RDS Mechanism Analysis 47 47. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 47. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q48	Q48: 48. A* Challenge: Arrhenius & RDS Mechanism Analysis 48 48. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 48. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q49	Q49: 49. A* Challenge: Arrhenius & RDS Mechanism Analysis 49 49. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 49. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]
Q50	Q50: 50. A* Challenge: Arrhenius & RDS Mechanism Analysis 50 50. (a) $\ln(1.5 \times 10^{-2} / 2.0 \times 10^{-4}) = \ln(75) = 4.317$ (1). $(1/300 - 1/350) = 4.76 \times 10^{-4} \text{ K}^{-1}$ (1). $E_a/R = 4.317 / 4.76 \times 10^{-4} = 9069 \text{ K}$ (1). $E_a = 9069 \times 8.31 = 75.4 \text{ kJ mol}^{-1}$ (1). 50. (b) Catalyst provides alternative route with lower E_a (1), increasing k (1).	[5]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 11A**FAQ #1: Units of Rate Constant k****Category: Calculation Trap****COMMON EXAMINER TRAP / MISTAKE:**

Using $\text{mol dm}^{-3} \text{ s}^{-1}$ for k regardless of overall order.

MODEL ANSWER / EXAMINER EXPECTATION:

Substitute units into rate equation: units of $k = (\text{mol dm}^{-3} \text{ s}^{-1}) / (\text{mol dm}^{-3})^n$. For 1st order: s^{-1} ; 2nd order: $\text{dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$; 3rd order: $\text{dm}^6 \text{ mol}^{-2} \text{ s}^{-1}$.

FAQ #2: Arrhenius Temperature in Kelvin**Category: Units Trap****COMMON EXAMINER TRAP / MISTAKE:**

Inserting temperature in Celsius into the Arrhenius equation.

MODEL ANSWER / EXAMINER EXPECTATION:

Always convert temperature to Kelvin ($K = ^\circ\text{C} + 273.15$). Using Celsius leads to negative or nonsense activation energy values.

FAQ #3: Activation Energy Units (kJ vs J)**Category: Conversion Trap****COMMON EXAMINER TRAP / MISTAKE:**

Forgetting to multiply/divide by 1000 when $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$ is used.

MODEL ANSWER / EXAMINER EXPECTATION:

R is given in Joules ($\text{J K}^{-1} \text{ mol}^{-1}$). The gradient $-E_a/R$ yields E_a in J mol^{-1} . Divide by 1000 to express final E_a in kJ mol^{-1} .

FAQ #4: Zero Order Reactants in Rate Equation**Category: Mechanism Trap****COMMON EXAMINER TRAP / MISTAKE:**

Including zero-order reactants in the rate equation.

MODEL ANSWER / EXAMINER EXPECTATION:

Zero-order reactants have rate $[X]^0 = 1$. Omit them from the rate equation completely, but note they participate after the RDS.

FAQ #5: Rate-Determining Step (RDS) Identification**Category: Theory Trap****COMMON EXAMINER TRAP / MISTAKE:**

Including species formed in fast steps AFTER the RDS in the rate equation.

MODEL ANSWER / EXAMINER EXPECTATION:

The rate equation only contains species present in or before the rate-determining step. Products of fast subsequent steps do not appear.

FAQ #6: Constant Half-Life Significance**Category: Graph Interpretation****COMMON EXAMINER TRAP / MISTAKE:**

Assuming constant half-life applies to zero or second order reactions.

MODEL ANSWER / EXAMINER EXPECTATION:

ONLY first-order reactions have a constant half-life independent of initial concentration ($t_{1/2} = \ln 2 / k$).

FAQ #7: Initial Rate Method Gradient Tangent**Category: Practical Skills****COMMON EXAMINER TRAP / MISTAKE:**

Drawing tangents at non-zero times when calculating initial rate.

MODEL ANSWER / EXAMINER EXPECTATION:

Initial rate MUST be determined by drawing a tangent at $t = 0$ seconds (when zero product has accumulated).

FAQ #8: Quenching Reaction Samples**Category: Practical Technique****COMMON EXAMINER TRAP / MISTAKE:**

Titration samples directly without quenching first.

MODEL ANSWER / EXAMINER EXPECTATION:

Quenching (ice-cold water/acid) halts or drastically slows the reaction so concentration does not change during titration.

FAQ #9: Catalysts in Rate Equations**Category: Homogeneous Catalysts****COMMON EXAMINER TRAP / MISTAKE:**

Excluding homogeneous catalysts from the rate equation.

MODEL ANSWER / EXAMINER EXPECTATION:

Homogeneous catalysts ARE involved in the rate-determining step and DO appear in the rate equation (e.g. H^+ in acid-catalysed esterification).

FAQ #10: Gradient of Arrhenius Plot**Category: Graph Analysis****COMMON EXAMINER TRAP / MISTAKE:**

Setting gradient equal to $-E_a$ instead of $-E_a/R$.

MODEL ANSWER / EXAMINER EXPECTATION:

On a plot of $\ln k$ against $1/T$, gradient = $-E_a / R$. Therefore $E_a = -\text{gradient} \times 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$.